ROBUST SIGNAL PROCESSING FOR WIRELESS SYSTEMS

Çağrı Şükrü ÖZDOĞAN

İzmir Institute of Technology
October 2026

ROBUST SIGNAL PROCESSING FOR WIRELESS SYSTEMS

A Thesis Submitted to
the Graduate School of Engineering and Sciences of
İzmir Institute of Technology
in Partial Fulfillment of the Requirements for the Degree of

DOCTOR OF PHILOSOPHY

in Electrical and Electronics Engineering

by
Çağrı Şükrü ÖZDOĞAN

October 2026
İZMİR

We approve the thesis of Çağrı Şükrü ÖZDOĞAN

Examining Committee Members:

Prof. Dr. Jüri Üyesi Numara 1 UZUNSOYADOĞLU
Afyon Kocatepe Üniversitesi Sağlık Bilimleri Fakültesi

Prof. Dr. Jüri Üyesi Numara 2 UZUNSOYADOĞLU
Afyon Kocatepe Üniversitesi Sağlık Bilimleri Fakültesi

Prof. Dr. Jüri Üyesi Numara 3 UZUNSOYADOĞLU
Afyon Kocatepe Üniversitesi Sağlık Bilimleri Fakültesi

Prof. Dr. Jüri Üyesi Numara 4 UZUNSOYADOĞLU
Afyon Kocatepe Üniversitesi Sağlık Bilimleri Fakültesi

Prof. Dr. Jüri Üyesi Numara 5 UZUNSOYADOĞLU
Afyon Kocatepe Üniversitesi Sağlık Bilimleri Fakültesi

10 October 2026

Prof. Dr. İlknur ÇİÇEK
Supervisor
Electrical and Electronics Engineering,
IZTECH

Assoc. Prof. Gökhan IŞIK
Co-Supervisor
Electrical and Electronics Engineering,
IZTECH

Prof. Dr. Ayşe YILMAZ
Head of the Department
Electrical and Electronics Engineering

Prof. Dr. Deniz KAYA
Dean of the Graduate School of
Engineering and Sciences

PREFACE

This thesis combines three related articles.

ACKNOWLEDGMENTS

I thank my supervisor and colleagues for their support.

ABSTRACT

ROBUST SIGNAL PROCESSING FOR WIRELESS SYSTEMS

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

ÖZET

KABLOSUZ SİSTEMLER İÇİN DAYANIKLI SİNYAL İŞLEME

Bu doğrulama örneğinde kablosuz sistemlerde sinyal işleme incelenmektedir. Ölçüm sonuçları kuramsal tahminlerle karşılaştırılmış; yöntem, bulgular ve sınırlılıklar açıklanmıştır. Türkçe karakterler: ÇĞİÖŞÜ çğıöşü.

TABLE OF CONTENTS

PREFACE..... iv
ACKNOWLEDGMENTS..... v
ABSTRACT..... vi
ÖZET..... vii
SYMBOLS AND ABBREVIATIONS..... ix
CHAPTER 1 1
INTRODUCTION
CHAPTER 2..... 2
ARTICLE 1
CHAPTER 3..... 3
ARTICLE 2
CHAPTER 4..... 4
ARTICLE 3
CHAPTER 5 5
CONCLUSIONS AND RECOMMENDATIONS
REFERENCES..... 6
APPENDIX A..... 7

SYMBOLS AND ABBREVIATIONS

SNR: Signal-to-noise ratio

CHAPTER 1

INTRODUCTION

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

CHAPTER 2

ARTICLE 1

Publication citation: Example article 1, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

CHAPTER 3

ARTICLE 2

Publication citation: Example article 2, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

CHAPTER 4

ARTICLE 3

Publication citation: Example article 3, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

REFERENCES

Smith, A. Signal Processing Methods; Example Press: İzmir, 2026.

APPENDIX A

Supplementary measurement protocol.

VITA

Çağrı Şükrü ÖZDOĞAN studied electrical engineering and researches signal processing.